

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

4.3 Continuous Random Variable

In sections 4.3.1 and 4.3.2, unless specified otherwise, let X, Y be two continuous r.v.'s defined on sample spaces $\mathbb{S}_X, \mathbb{S}_Y \in \mathbb{R}$ respectively.

4.3.1 Characterization

Definition 4.3.1.1. The probability distribution or **probability density function** (pdf) of X , denoted as $f_X(\cdot)$, is a function such that for any $a, b \in \mathbb{R}$ so that $a \leq b$,

$$P(X \in [a, b]) = \int_a^b f_X(x) dx.$$

- In other words, pdf gives the probability of every event constituted by an interval on the real line.
- A pdf $f_X(\cdot)$ is well-defined iff it satisfies the following two conditions:
 - $f_X(x) \geq 0$ for all $x \in \mathbb{R}$.
 - $\int_{-\infty}^{+\infty} f_X(x) dx = 1$.
- By Definition 4.3.1.1., $P(X = a) = 0$ for all $a \in \mathbb{R}$. Thus, for any $a, b \in \mathbb{R}$ so that $a \leq b$,

$$\int_a^b f_X(x) dx = P(X \in [a, b]) = P\{X \in (a, b]\} = P\{X \in [a, b)\} = P[X \in (a, b)].$$

Definition 4.3.1.2. The **cumulative distribution function** (CDF) of X , denoted as $F_X(\cdot)$, is given by

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(z) dz \text{ for all } x \in \mathbb{R}.$$

- Recall that a CDF $F_X(\cdot)$ is well-defined iff it satisfies the following three conditions:
 - $F_X(x) \leq F_X(x')$ for all $x, x' \in \mathbb{R}$ so that $x < x'$.
 - $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
 - $\lim_{x \downarrow x_0} F_X(x) = F_X(x_0)$, that is, $F_X(\cdot)$ is right-continuous - particularly for the case of continuous r.v.'s, $F_X(\cdot)$ is continuous.

Proposition 4.3.1.3. For any $a, b \in \mathbb{R}$ and $a \leq b$,

$$P(X > a) = 1 - F_X(a) \text{ and } P(x \in [a, b]) = F_X(b) - F_X(a).$$

- Proof idea: Direct calculation by definitions.

Proposition 4.3.1.4. For any $x \in \mathbb{R}$ so that the derivative $F'_X(x)$ exists,

$$F'_X(x) = f_X(x).$$

- Proof idea: Fundamental theorem of calculus.

Definition 4.3.1.5. The **joint probability density function** of X and Y - or (X, Y) - denoted as $f_{XY}(\cdot, \cdot)$, is a function such that for any two-dimensional set $A \subset \mathbb{R}^2$,

$$P[(X, Y) \in A] = \iint_A f_{XY}(x, y) dx dy.$$

Definition 4.3.1.6. The **marginal probability density function** of X , denoted as $f_X(\cdot)$, is given by

$$f_X(x) = \int_{-\infty}^{+\infty} f_{XY}(x, y) dy.$$

- The manner in which we defined families of probability distributions of discrete r.v.'s in Definition 4.2.1.4 also applies to continuous r.v.'s.

Definition 4.3.1.7. X and Y are said to be **independent** if

$$f_{XY}(x, y) = f_X(x)f_Y(y) \text{ for all } x, y \in \mathbb{R}$$

and are said to be dependent if otherwise, i.e., $f_{XY}(x, y) \neq f_X(x)f_Y(y)$ for some $x, y \in \mathbb{R}$.

4.3.2 Statistics

Definition 4.3.2.1. The **expected value**, or mean value, of X , denoted as $E(X)$ - or μ_X (or just μ in univariate context) - is

$$E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx.$$

Definition 4.3.2.2. The q^{th} **percentile** of X is the value of x so that - in the context of continuous r.v., where $F_X(\cdot)$ is normally continuous,

$$F_X(x) = q\%.$$

- Recall the special cases of percentile: median when $q = 50$, and quartiles when $q = 25$ or $q = 75$.

Definition 4.3.2.3. The **variance** of X , denoted as $V(X)$ - or σ_X^2 (or just σ^2 in univariate context) - is

$$V(X) = \int_{-\infty}^{+\infty} [x - E(X)]^2 f_X(x) dx = E\{[X - E(X)]^2\},$$

and the **standard deviation** of X is $\sigma_X = \sqrt{V(X)}$.

Definition 4.3.2.4. The **covariance** between X and Y , denoted as $Cov(X, Y)$, is

$$Cov(X, Y) = E\{[X - E(X)][Y - E(Y)]\} = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} [x - E(X)][y - E(Y)] f_{XY}(x, y) dx dy.$$

- The following things that we have stated in the context of discrete r.v. in section 4.2 can be directly extended to cases with continuous r.v.'s:
 - Alternative formulas of variance (Proposition 4.2.2.4) and covariance (Proposition 4.2.2.8).
 - Definitions of correlation (Definition 4.2.2.9.1) and uncorrelated (Definition 4.2.2.11).
 - Propositions related to particular values of correlation (Proposition 4.2.2.10 and Proposition 4.2.2.12).
 - Definitions of linear transformation (Definition 4.2.2.5) and linear combination (Definition 4.2.2.14).
 - Expectation, median (percentile), variance, covariance, and correlation under linear transformations (Proposition 4.2.2.6 and Proposition 4.2.2.13).
 - Expectation and covariance (and subsequently, variance and correlation) under linear combinations (Proposition 4.2.2.15).

4.3.3 Uniform Distribution

Definition 4.3.3.1. A continuous r.v. X is said to follow a **uniform distribution** on an interval $[a, b] \subset \mathbb{R}$, denoted as $X \sim U[a, b]$, if its pdf is

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in [a, b] \\ 0 & \text{elsewhere} \end{cases}.$$

- For $X \sim U[a, b]$,

– its CDF is

$$F_X(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } x \in (a, b] \\ 1 & \text{if } x > b \end{cases},$$

- $E(X) = \frac{a+b}{2}$, and $V(X) = \frac{(a-b)^2}{12}$ - both of which are from direct calculation by definitions.

4.3.4 Normal Distribution

Definition 4.3.4.1. A r.v. X is said to follow a **normal distribution** with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$, denoted as $X \sim N(\mu, \sigma^2)$, if its pdf is

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \text{ for all } x \in \mathbb{R}.$$

- For $X \sim N(\mu, \sigma^2)$,
 - its pdf is symmetric around μ ,
 - $E(X) = m(X) = \mu$, which is straightforwardly from symmetry,
 - $V(X) = \sigma^2$, which makes use of the key fact that $\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$, and
 - (3- σ rule of thumb) $P(X \in [\mu - \sigma, \mu + \sigma]) \approx 0.68$, $P(X \in [\mu - 2\sigma, \mu + 2\sigma]) \approx 0.95$, and $P(X \in [\mu - 3\sigma, \mu + 3\sigma]) \approx 0.997$.

Definition 4.3.4.2. A r.v. X is said to follow a **standard normal distribution** if $X \sim N(0, 1)$.

- A standard normal r.v. is conventionally denoted as Z , with its CDF conventionally denoted as $\Phi(\cdot)$.
- The $100(1 - \alpha)^{th}$ percentile of Z is conventionally denoted as z_α - and such z_α 's are usually referred to as z critical values.
 - Such critical values of specific distributions are frequently used in inferential statistics.

Proposition 4.3.4.3. $X \sim N(\mu, \sigma^2)$ iff

$$\frac{X - \mu}{\sigma} = Z \text{ - and thus } F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

- Proof idea: Integration by change of variable.

Corollary 4.3.4.4. A linear transformation of a normal r.v. is still a normal r.v..

- Proof idea: Directly by Proposition 4.3.4.3.

Proposition 4.3.4.5. A linear combination of two independent normal r.v.'s is still a normal r.v..

- Proof: not required.
 - For your reference, one rather straightforward proof idea makes use of the key fact that two independent normal r.v.'s are jointly normal, and then apply the vector/matrix version of Corollary 4.3.4.4.
- Proposition 4.3.4.5 straightforwardly implies that a linear combination of finitely many independent normal r.v.'s is still a normal r.v..
- Proposition 4.3.4.5 still holds even if the r.v.'s are not independent.

4.3.5 Chi-Squared Distribution

Definition 4.3.5.1. Let $Z_1, Z_2, \dots, Z_k$ be $k \in \mathbb{N}_+$ independent and identically distributed (i.i.d.) standard normal r.v.'s, then a r.v. X is said to follow a **chi-squared distribution** with k degrees of freedom, denoted as $X \sim \chi_k^2$, if

$$X = \sum_{i=1}^k Z_i^2.$$

- The pdf of χ_k^2 extends from 0 to $+\infty$, is positively skewed, and becomes flatter as well as more symmetric as k increases.
- $E(\chi_k^2) = k$ and $V(\chi_k^2) = 2k$ - proof not required.
- The $100(1 - \alpha)^{th}$ percentile of χ_k^2 is conventionally denoted as $\chi_{\alpha,k}^2$ - and such $\chi_{\alpha,k}^2$'s are usually referred to as chi-squared critical values.

4.3.6 t distribution

Definition 4.3.6.1. Let $Z \sim N(0, 1)$ and $K \sim \chi_k^2$ be independent, then a r.v. X is said to follow a **(student) t distribution** with k degrees of freedom, denoted as $X \sim t_k$, if

$$X = \frac{Z}{\sqrt{\frac{\chi_k^2}{k}}}.$$

- The pdf of χ_k^2 extends from $-\infty$ to $+\infty$, symmetric around 0, bell-shaped, and converges to Z as $k \rightarrow +\infty$.
 - Correspondingly, Z could be considered as a t distribution with ∞ degrees of freedom.
- $E(t_k) = 0$ (by symmetry) and $V(t_k) = \frac{k}{k-2} > 1$ - proof not required.
 - Corresponding to $V(t_k) > V(Z) = 1$, the pdf of a t distribution is more spread out (i.e., with fatter tails) than that of Z .
- The $100(1 - \alpha)^{th}$ percentile of t_k is conventionally denoted as $t_{\alpha,k}$ - and such $t_{\alpha,k}$'s are usually referred to as t critical values.

Example 4.A

Consider a college instructor who barely ends his class on time and may delay by X minutes. Suppose the delay X follow a probability distribution characterized by the following pdf:

$$f_X(x) = \begin{cases} x & \text{if } x \in [0, 1] \\ 2 - x & \text{if } x \in (1, 2] \\ 0 & \text{else} \end{cases}.$$

1. Graph the above pdf. Is it well-defined?
2. How much is the probability that the instructor delays for exactly 1 minute?
3. How much is the probability that the delay is longer than 1.5 minutes?
4. How much is the probability that the delay is between 0.5 - 1.5 minutes?
5. How much is the probability that the delay is no more than y minutes? Graph it as a function of y .
6. How much is the expectation (average) of the duration of delay (unit: minute)?
7. How much is the 25% percentile of the duration of delay (unit: minute)?
8. How much is the variance of the duration of delay (unit: minute)?

Example 4.B

Consider a sprinter who can finish a 100-meter sprint in Y seconds, where $Y \sim U[9.8, 10.6]$.

1. Write down the pdf of Y and graph it.
2. How much is the probability that the sprinter finishes the sprint within 10 seconds?
3. Work out the CDF of Y and graph it.
4. How much is the expected time for the sprinter to finishes the sprint (unit: second)?
5. How much is the median of the time for the sprinter to finishes the sprint (unit: second)?
6. How much is the standard error of the time for the sprinter to finishes the sprint (unit: second)?

Example 4.C

Consider a standardized test that is used for evaluations in a university. Students' scores on this test are approximately normally distributed, with mean 500 and standard deviation 100.

1. How much is the probability that a randomly-chosen student has a score of 696 or higher?
2. Suppose that the university wants to identify the 95% percentile of the grade distribution so that it can give scholarships to its top 5% students. How much is the 95% percentile of the grade distribution?
3. Suppose that the university is interested in the performance of the 99% students in the middle of the grade distribution. Find out an interval that is symmetric around the average grade and includes 99% of the scores.

Note: We will need the following z critical values in this problem: $z_{0.05} = 1.6449$, $z_{0.025} = 1.96$, $z_{0.01} = 2.3263$, $z_{0.005} = 2.5758$.

Example 4.D

Suppose John Doe runs a business that earns him $X_1 \sim N(6, 0.49)$ hundred dollars per week, and his wife Jane Doe also runs a business that earns her $X_2 \sim N(4, 0.16)$ hundred dollars per week.

1. How much is the average total weekly income of the Doe couple (unit: hundred dollar)?
2. Count a month as 4 weeks throughout this problem. How much is the average total monthly income of the Doe couple (unit: hundred dollar)?
3. Suppose throughout the rest of the problem that the covariance between the weekly incomes of the couple is -0.2 (unit: hundred dollar). How much is the variance of the total weekly income of the Doe couple (unit: hundred dollar)?
4. How much is the variance of the total monthly income of the Doe couple (unit: hundred dollar)?
5. How much is the correlation between the weekly incomes of the couple (unit: hundred dollar)?
6. How much is the correlation between the monthly incomes of the couple (unit: hundred dollar)?
7. How much is the 10% percentile of the total weekly income of the Doe couple (unit: hundred dollar)?
8. How much is the 10% percentile of the total monthly income of the Doe couple (unit: hundred dollar)?

Note: We will need the following z critical values in this problem: $z_{0.1} = 1.2816$.